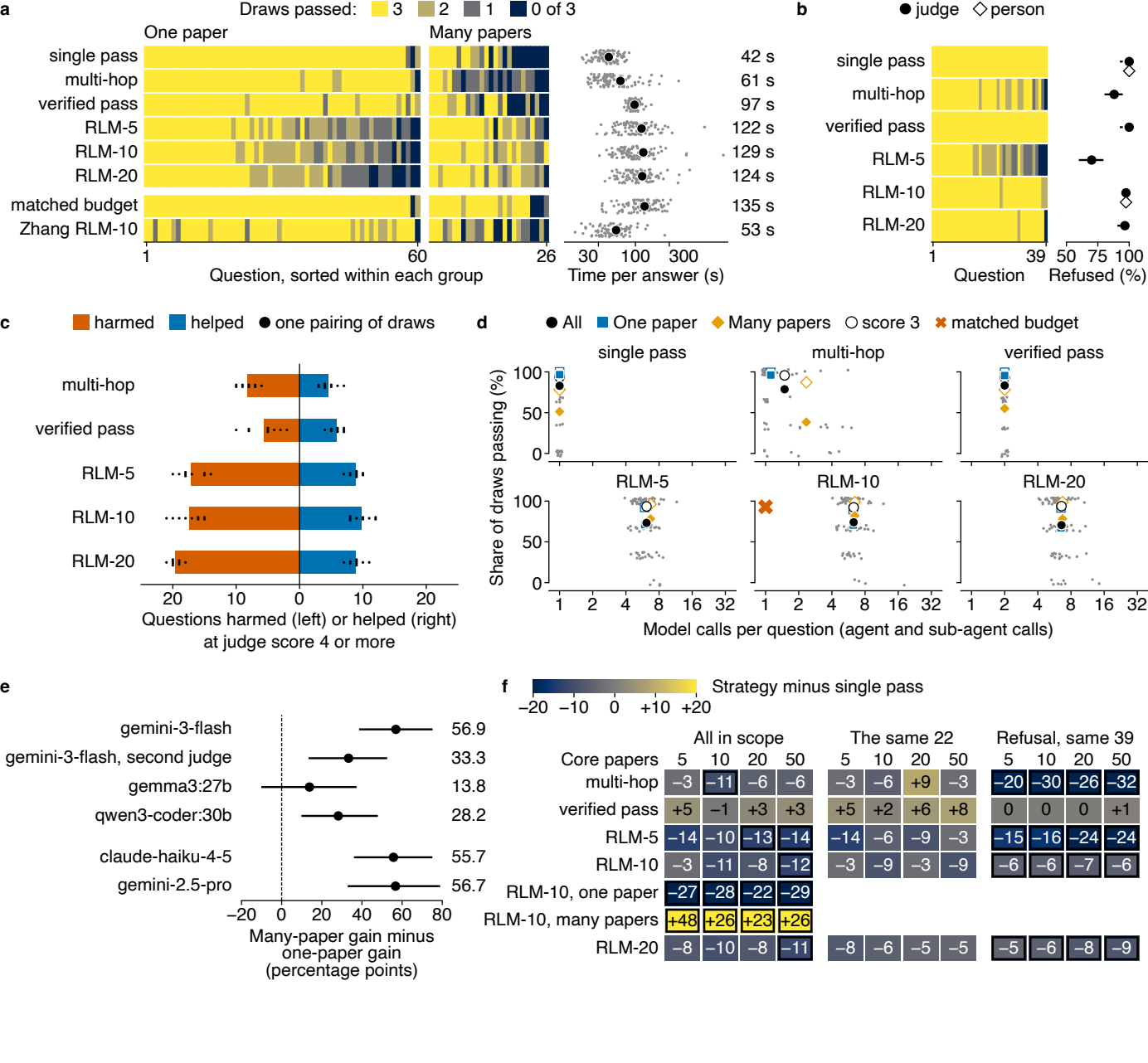

a
Draws passed: 3 2 1 0 of 3
One paper
Many papers
single pass
multi-hop
verified pass
RLM-5
RLM-10
RLM-20
matched budget
Zhang RLM-10
42 s
61 s
97 s
122 s
129 s
124 s
135 s
53 s
1
60
26
30 100 300
Question, sorted within each group
Time per answer (s)
b
judge person
single pass
multi-hop
verified pass
RLM-5
RLM-10
RLM-20
1
39
50 75 100
Question
Refused (%)
c
harmed helped one pairing of draws
multi-hop
verified pass
RLM-5
RLM-10
RLM-20
20 10 0 10 20
Questions harmed (left) or helped (right)
at judge score 4 or more
d
All One paper Many papers score 3 matched budget
single pass
multi-hop
verified pass
RLM-5
RLM-10
RLM-20
Share of draws passing (%)
100
50
0
1 2 4 8 16 32
Model calls per question (agent and sub-agent calls)
e
gemini-3-flash 56.9
gemini-3-flash, second judge 33.3
gemma3:27b 13.8
qwen3-coder:30b 28.2
claude-haiku-4-5 55.7
gemini-2.5-pro 56.7
−20 0 20 40 60 80
Many-paper gain minus
one-paper gain
(percentage points)
f
Strategy minus single pass
−20 −10 0 +10 +20
All in scope
The same 22
Refusal, same 39
Core papers 5 10 20 50 5 10 20 50 5 10 20 50
multi-hop −3 −11 −6 −6 −3 −6 +9 −3 −20 −30 −26 −32
verified pass +5 −1 +3 +3 +5 +2 +6 +8 0 0 0 +1
RLM-5 −14 −10 −13 −14 −14 −6 −9 −3 −15 −16 −24 −24
RLM-10 −3 −11 −8 −12 −3 −9 −3 −9 −6 −6 −7 −6
RLM-10, one paper −27 −28 −22 −29
RLM-10, many papers +48 +26 +23 +26
RLM-20 −8 −10 −8 −11 −8 −6 −5 −5 −5 −6 −8 −9